# Urban Solar Carver: A GPU-Accelerated Framework for Solar-Driven Urban Form Finding

Aristotelis Vartholomaios(✉)

University of Thessaly, Volos 383 34, Greece
avartholomaios@uth.gr

**Abstract.** From early zoning codes to contemporary environmental standards, building volume regulation has served as a primary mechanism for guiding urban form and protecting access to sun and light. The study presents the Urban Solar Carver (USC), a computational framework for generating maximum developable building volumes that protect solar and daylight access. USC builds upon earlier works on solar voxel-based form-finding methods, embedding them into a unified, transparent and reproducible framework. Its main advantage is the integration of GPU-accelerated geometry voxelization, raytracing and solar and sky radiation modelling to evaluate millions of ray–surface–voxel interactions at near real-time speeds. USC is currently under development as a standalone Python library with its own command line interface and connectors to Grasshopper for Rhinoceros 3d. It can generate conventional daylight and time-based solar envelopes but can also perform advanced volume carving using performance-based criteria such as cumulative solar irradiation, passive solar benefit, daylighting benefit, outdoor comfort and even radiative cooling benefit. Some of these features are stable while others are still experimental and require further validation. USC utilizes multiple thresholding techniques, moving beyond fixed-rule envelopes and toward performance-responsive form finding. This level of interaction lets designers and planners experiment with different constraints and immediately see how these choices shape the final massing.

**Keywords:** solar access · urban planning · urban form · building volume · performance-based design

## 1 Introduction

Access to sunlight and daylight has long influenced the regulation of urban form [1, 2]. Geometric constraints such as setbacks, height-to-street-width ratios, cut-off planes, and solar access rules are frequently employed to limit overshadowing and maintain adequate environmental conditions for buildings and public space [3]. Interest in solar access has increased in the context of climate-responsive urbanism, where solar exposure affects daylight availability, passive heating, outdoor comfort, and local renewable energy use [4].

© The Author(s) 2026
L. Bragança et al. (Eds.): ReSBE 2026, LNCE 905, pp. 600–607, 2026.
https://doi.org/10.1007/978-3-032-34930-9_70

Urban solar-access regulation has gradually evolved from simple geometric rules toward solar-informed volumetric controls. Early approaches relied on setbacks, height-to-street-width ratios, and later sky-exposure planes, as in New York's 1916 zoning framework, to limit building mass [2]. The solar envelope concept introduced in the 1970s marked a shift from fixed angular restrictions to buildable volumes derived from sun-path geometry and protected periods of solar access [5]. Subsequent work expanded this lineage through methods such as solar fans [6], the Residential Solar Block envelope [7], iso-solar surfaces [8], reverse or subtractive solar envelopes [9], and solar-collection envelopes [10]. Digital envelope generation methods have been developed since the 1990s ranging from raster and point-grid approximations [6, 11] to Boolean solid operations [7] and voxel-based approaches [9, 12, 13], each involving trade-offs in precision, robustness, speed, and usability. Some tools implement only a single envelope logic, while others are tied to proprietary software environments or narrow workflows [6]. As solar-access ambitions expand, practical tools for planners and designers remain limited.

This paper introduces the Urban Solar Carver (USC), a computational framework for producing maximum developable urban volumes. USC follows a subtractive logic: a theoretical maximum volume is progressively carved to satisfy environmental constraints. The framework integrates GPU-accelerated voxelization, ray-based volumetric carving, solar and sky radiation modelling, and threshold-based filtering within a single workflow.

## 2 Tool Description

USC is implemented as a standalone Python library with a command line interface and connectors to Grasshopper for Rhinoceros 3D, enabling both reproducible analysis and interactive design use. A user-defined maximum candidate volume is first discretized into a three-dimensional voxel grid representing potential built mass at any scale (building, block, or neighborhood). The user also defines test surfaces requiring solar or sky access, such as windows, PV arrays, or entire façades. Surface normals must face outward. Rays are traced from sampled points on these surfaces toward multiple directions. Depending on the evaluation mode, these directions correspond to fixed angles, sun positions, or sky patches. Voxels intersected by the rays accumulate either violation counts or weighted obstruction scores. A user-defined threshold is then applied to the resulting voxel score field to determine which voxels are retained.

The workflow consists of three stages: preprocessing, thresholding, and exporting. Preprocessing includes geometry loading, voxelization of the candidate volume, sampling of test surfaces, generation of directional vectors, and per-voxel score computation. Thresholding converts the score field into a binary retention mask using fixed or automatic criteria. Exporting reconstructs the retained voxels into a polygonal mesh with optional cleanup or smoothing (Fig. 1). Parameters are controlled through a YAML configuration file, with optional runtime overrides for batch studies. The framework requires three minimum inputs: a maximum volume mesh, a test-surface mesh, and for climate-based analyses an EPW weather file. The candidate mass is voxelized at a user-defined resolution and test surfaces are sampled into point–normal pairs, controlling the trade-off between geometric fidelity and computational cost.

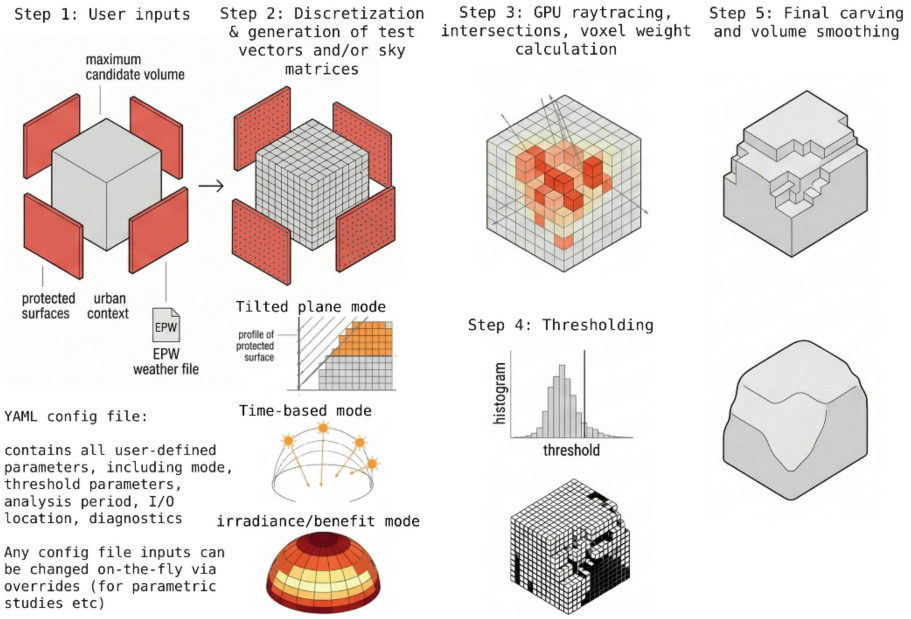



**Fig. 1** Overview of USC inputs, modes and execution steps.

Directional data are derived in two ways. For time-explicit analyses, Ladybug generates sun vectors for specified dates and hours. For climate-based analyses, USC uses weighted sky patches derived from annual weather data mapped onto the Tregenza sky subdivision. These create two analysis families: sun-vector methods, which evaluate obstruction at specific times, and sky-patch methods, which evaluate cumulative environmental opportunity over the sky hemisphere. In this study the demonstrated modes are tilted plane, time-based, irradiance, daylight, and benefit.

In *tilted plane* mode, USC applies an angular plane restriction defined by user-specified angles, optionally varying by orientation. Rays cast from sampled surfaces accumulate voxel violation counts according to this directional field. In *time-based* mode, sun vectors for a specified analysis period are generated and rays are cast toward these positions. Intersected voxels accumulate solar-access violations. With a zero-tolerance threshold the resulting mass approximates a conventional solar envelope. In *irradiance mode*, sky directions are weighted by cumulative annual radiation derived from EPW data. Voxels blocking directions with higher radiation receive higher scores and are preferentially removed. In *daylight mode*, sky directions are weighted using a CIE overcast sky luminance distribution. Voxels obstructing brighter sky regions accumulate higher scores. The *benefit mode* weights sky patches according to passive solar usefulness rather than radiation magnitude alone.

Weights are derived from the Ladybug *benefit sky matrix*, generated using Radiance's gendaymtx, which assigns a positive or negative weight to each Tregenza sky patch depending on whether radiation from that direction offsets heating loads or increases cooling loads, given temperature data from the EPW file [14]. The balance temperature

and balance offset parameters control these assumptions and can be refined through building energy simulation.

The computational core of USC evaluates large numbers of ray–voxel intersections. For each sampled surface point, a ray is cast toward each direction in the active set (sky-patch centre or sun position); the total ray count equals the product of test points and directional vectors. USC relies on NVIDIA Warp [15] for sparse voxel representation and ray tracing and on Ladybug Tools [14] for sun-position calculation and climate-based environmental weighting from EPW data. The framework can run on CPU or CUDA-enabled GPUs. Warp provides voxel and raytracing operations while USC handles carving logic, score accumulation, and data flow.

Preprocessing produces either a violation-count field (tilted plane, time-based) or a weighted obstruction field (irradiance, daylight, benefit). During thresholding, weighted fields may be normalized and converted into retention masks using user-defined thresholds or automatic methods such as Otsu or head/tail classification. A retained-fraction strategy can also be used to preserve a specified share of the score distribution. After thresholding, the retained voxel mask is reconstructed into a polygonal mesh with optional smoothing. Intermediate score fields and metadata are stored to enable selective reruns without repeating the full computation.

## 3 Methodology

USC is applied to a real neighborhood to test its capability to generate and compare alternative solar-driven urban volume controls within the same urban setting. The case study site is in western Thessaloniki, Greece which has Mediterranean climate (40°40′33.4″N, 22°54′47.9″E). This 15 ha neighborhood comprises mainly multifamily apartment buildings.

Building footprints and height data were extracted from OpenStreetMap and used to reconstruct the existing built fabric as a three-dimensional urban geometry (reference case). The same footprints were extruded to a common upper height limit to generate candidate redevelopment masses, retaining the parcel structure while providing a shared geometric basis for comparison. All vertical facade surfaces of the existing buildings were used as test surfaces.

Three USC modes were tested: (i) a fixed-plane scenario using a 45° angular restriction, (ii) a time-based solar-envelope scenario for the winter solstice from 10:00 to 14:00. And (iii) a benefit-based scenario guided by a threshold of 80% useful insolation. Climatic data were provided in EPW format from the IWEC database. For the benefit scenario balance temperature was set to 20 °C with a range of ±2 °C. The value was determined through a simple shoebox simulation of a residential mid-floor apartment using EnergyPlus [16] with construction and occupancy matching the Greek building energy code (KEvAK) specifications. Heating period according to the code for the respective climatic zone (Γ') is from 15$^{th}$ of October to 30$^{th}$ of April. For each scenario, USC carved the candidate masses and generated a final urban volume. These were compared with the existing built form, resulting in four examined cases: existing, fixed plane, time-based, and benefit. All four cases were then subjected to the same ex-post heating period solar simulation.

The comparison used two groups of indicators. Development intensity was described through total built volume, volume per site area, and an estimated mixed Floor Area Ratio (FAR) proxy, assuming an average floor-to-floor height of 3.5 m. Heating period solar performance was described through: (i) total cumulative insolation across all building envelopes and (ii) mean cumulative insolation in kWh/m$^2$, calculated as the average cumulative energy received per evaluated surface over the examined period. The latter was used as the main solar-performance indicator.

## 4 Results

Table 1 summarizes volumetric and solar-performance results for the four cases while Fig. 2 visualizes them. The existing neighborhood has a total built volume of 780,400 m$^3$, corresponding to a volumetric density of 5.08 m$^3$/m$^2$ and a mixed FAR of 1.45. Mean cumulative winter insolation is 214.76 kWh/m$^2$. Calculation time was 47 s per run on a workstation with 64 GB RAM and a GeForce RTX 3060 (12 GB VRAM) for 223 volumes and 1338 test surfaces at a voxel and grid resolution of 0.5 m (over 99 million voxels and 532,581 test points). With 145 Tregenza sky patches the total ray workload exceeded 77 million patch-test point queries per run.

The 45° fixed-plane scenario produced the largest envelope volume (Table 1), representing a 12.1% increase relative to the existing built volume. Mean cumulative insolation increased by +10.9% relative to existing. This scenario also yielded the highest total cumulative insolation due to its larger exposed surface area. The time-based scenario was the most restrictive, reducing total volume approximately 25.7% below the existing built volume (Table 1). Despite this reduction, mean cumulative insolation increased to about 23.5% above the existing condition. The benefit-based scenario nearly matched the volume of the fixed-plane case while achieving the highest mean cumulative winter insolation (Table 1). This represents increases of 34.7% relative to the existing neighborhood, 21.4% relative to the fixed-plane scenario, and 9.0% relative to the time-based case.

The three scenarios reveal clear density–solar trade-offs. The fixed-plane rule retains the largest buildable volume, the time-based envelope secures stronger winter solar

**Table 1.** Volumetric and winter solar performance comparison of the examined cases

| Scenario | Volume (m$^3$) | Volume/site area (m$^3$/m$^2$) | Mixed FAR | Mean cumulative winter insolation (kWh/m$^2$) | Total cumulative insolation (GWh) |
|---|---|---|---|---|---|
| Existing | 780,400 | 5.0 | 1.4 | 214 | 57.15 |
| Fixed plane | 874,687 | 5.7 | 1.6 | 238 | 62.54 |
| Time-based | 579,806 | 3.8 | 1.0 | 265 | 56.01 |
| Benefit | 869,904 | 5.7 | 1.6 | 289 | 60.58 |

access at the cost of density, and the benefit-based method achieves the highest mean solar performance while maintaining a volume comparable to the fixed-plane case.

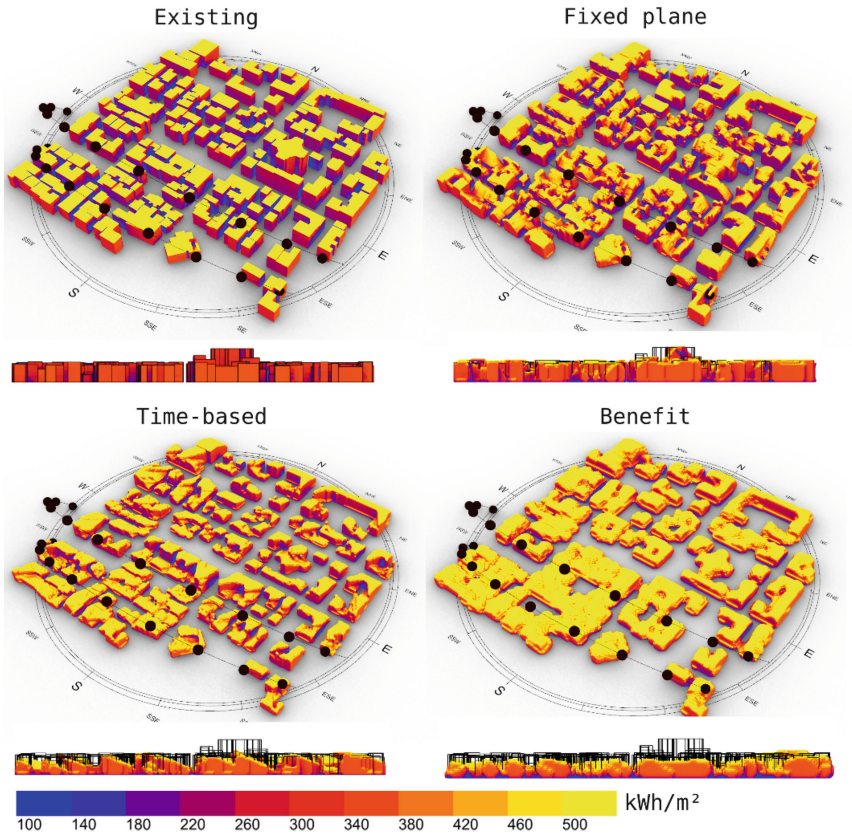



**Fig. 2.** Average cumulative insolation for the heating period for the four examined cases. Indicative cross-sections illustrate the volumetric differences between the current built form and the three generated envelopes. Sun positions for start and end dates shown as black dots.

## 5 Conclusions

This paper presented USC, a voxel-based GPU-accelerated framework for solar-driven urban form finding. The framework integrates different volumetric control logics within a common computational pipeline and generates redevelopment envelopes in a reproducible manner. In the neighborhood-scale experiment, three control modes were tested: a 45° fixed-plane rule, a winter-solstice time-based solar envelope, and a benefit-weighted solar criterion. The scenarios produced markedly different density–solar outcomes with the benefit-based scenario retaining nearly the same volume as the fixed-plane case while producing the highest mean cumulative winter insolation among all tested cases.

The analysis is exploratory and limited to a single neighborhood, while the benefit mode relies on simplified assumptions regarding passive-solar usefulness. Despite these limitations, USC can serve as a general computational basis for subtractive urban form finding, allowing different environmental constraints to be compared within the same framework and supporting future research on climate-responsive volumetric controls.

**Acknowledgements.** The research is conducted in the operating framework of the University of Thessaly Innovation, Technology Transfer Unit and Entrepreneurship Center "One Planet Thessaly", under the "University of Thessaly Grants for Scientific Publication Support" action and is funded by the Special Account of Research Grants of the University of Thessaly.

## References

1. Littlefair, P.: Passive solar urban design: ensuring the penetration of solar energy into the city. Renew. Sust. Energ. Rev. **2**(3), 303–326 (1998). https://doi.org/10.1016/S1364-0321(97)00009-9
2. DeKay, M., Brown, G.Z.: Sun, Wind & Light Architectural Design Strategies. John Wiley & Sons (2014)
3. Freitas, S., Catita, C., Redweik, P., Brito, M.C.: Modelling solar potential in the urban environment: state-of-the-art review. Renew. Sust. Energ. Rev. **41**, 915–931 (2015). https://doi.org/10.1016/j.rser.2014.08.060
4. Natanian, J., Kastner, P., Dogan, T., Auer, T.: From energy performative to livable Mediterranean cities: an annual outdoor thermal comfort and energy balance cross-climatic typological study. Energy Build. **224**, 110283 (2020). https://doi.org/10.1016/j.enbuild.2020.110283
5. Knowles, R.L.: The solar envelope: its meaning for energy and buildings. Energy. **35**, 15–25 (2003)
6. De Luca, F., Dogan, T.: A novel solar envelope method based on solar ordinances for urban planning. Build. Simul. **12**(5), 817–834 (2019). https://doi.org/10.1007/s12273-019-0561-1
7. Vartholomaios, A.: The residential solar block envelope: a method for enabling the development of compact urban blocks with high passive solar potential. Energy Build. **99**, 303–312 (2015). https://doi.org/10.1016/j.enbuild.2015.04.046
8. Ratti, C., Morello, E.: SunScapes: extending the 'solar envelopes' concept through 'iso-solar surfaces' SunScapes: extending the 'solar envelopes' concept through 'iso-solar surfaces.'. In: PLEA2005 - the 22nd Conference on Passive and Low Energy Architecture, pp. 1–6 (2005)
9. De Luca, F., Dogan, T., Sepúlveda, A.: Reverse solar envelope method. A new building form-finding method that can take regulatory frameworks into account. Autom. Constr. **123**, 103518 (2021). https://doi.org/10.1016/j.autcon.2020.103518
10. Capeluto, I.G., Shaviv, E.: On the use of 'solar volume' for determining the urban fabric. Sol. Energy. **70**(3), 275–280 (2001). https://doi.org/10.1016/S0038-092X(00)00088-8
11. Morello, E., Gori, V., Balocco, C., Ratti, C.: Sustainable urban block design through passive architecture: a tool that uses urban geometry optimization to compute energy savings. In: Proceedings of PLEA2009 -26th Conference on Passive and Low Energy Architecture-PLEA2009 - 26th Conference on Passive and Low Energy Architecture, (June), pp. 22–24 (2009)
12. Alkadri, M.F., De Luca, F., Turrin, M., Sariyildiz, S.: A computational workflow for generating a voxel-based design approach based on subtractive shading envelopes and attribute information of point cloud data. Remote Sens. **12**(16), 2561 (2020). https://doi.org/10.3390/rs12162561

13. Natanian, J., Luca, F.D., Wortmann, T., Capeluto, G.: The solar block generator: an additive parametric method for solar driven urban block design. J. Phys. Conf. Ser. **2042**(1), 012049 (2021). https://doi.org/10.1088/1742-6596/2042/1/012049
14. Roudsari, M.S., Pak, M., Smith, A.: Ladybug: a parametric environmental plugin for grasshopper to help designers create an environmentally-conscious design. In: 13th Conference of International Building Performance Simulation Association, pp. 3129–3135 (2013)
15. Macklin, M.: Warp: A High-Performance Python Framework for GPU Simulation and Graphics [Python]. NVIDIA Warp (2022) https://github.com/NVIDIA/warp. (Original work published 2022)
16. Crawley, D.B. et al.: EnergyPlus: creating a new-generation building energy simulation program. Energy Build. **33**(4), 319–331 (2001). https://doi.org/10.1016/S0378-7788(00)00114-6

**Open Access** This chapter is licensed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits any noncommercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if you modified the licensed material. You do not have permission under this license to share adapted material derived from this chapter or parts of it.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

